

Chapter 1: Computer Skills

Class 1 – Computer Fundamentals, Hardware Devices & Computer System

Learning Objectives

After completing this lesson, students will be able to:

- Understand the basic concepts of computers.
 - Identify the main parts of a computer system.
 - Differentiate between hardware and software.
 - Recognize internal and external hardware devices.
 - Explain the functions of common computer hardware.
 - Understand the importance and uses of computers in everyday life.
-

Introduction

A **computer** is an electronic device that accepts data, processes it, stores information, and produces useful results. Computers help us perform many tasks quickly, accurately, and efficiently. Today, computers are used in schools, homes, hospitals, banks, offices, shops, and many other places.

Understanding computer fundamentals helps students develop digital skills that are useful for learning and everyday life.

What is a Computer?

A **computer** is an electronic machine that:

- Accepts data (Input)
- Processes the data
- Stores information
- Produces useful output

Simple Definition

Computer = Input → Process → Output → Storage

Examples of Computer Use

- Studying online
 - Playing educational games
 - Drawing pictures
 - Typing documents
 - Watching educational videos
 - Searching for information
 - Communicating with others
-

Computer Fundamentals

Computer fundamentals are the basic concepts required to understand how a computer works.

They include:

- Computer System
 - Hardware
 - Software
 - Input Devices
 - Output Devices
 - Storage Devices
 - Operating System
-

What is a Computer System?

A **computer system** is made up of different parts that work together to perform tasks.

Main Components of a Computer System

- Hardware
- Software
- User
- Data

Each component is important for the computer to function properly.

What is Hardware?

Hardware refers to all the physical parts of a computer that we can see and touch.

Examples

- Monitor
 - CPU (System Unit)
 - Keyboard
 - Mouse
 - Printer
 - Speakers
 - Scanner
 - Webcam
-

Types of Hardware

1. Internal Hardware

Internal hardware is located **inside** the computer cabinet.

Examples

- Motherboard
- Processor (CPU)
- RAM (Memory)
- Hard Disk / SSD
- Power Supply Unit

Functions

- Processes data
 - Stores information
 - Controls computer operations
 - Supplies power to components
-

2. External Hardware

External hardware is connected to the computer from the outside.

Examples

- Monitor
- Keyboard
- Mouse
- Printer
- Speakers
- Scanner
- Webcam
- Microphone

Functions

- Allows users to enter data
- Displays information
- Prints documents
- Produces sound
- Captures images and audio

Functions of Common Hardware Devices

Hardware Device	Function
Monitor	Displays text, images, and videos.
Keyboard	Used to type letters, numbers, and symbols.
Mouse	Controls the pointer and selects items.
CPU	Processes instructions and controls the computer.
Printer	Prints documents and pictures.
Scanner	Converts paper documents into digital form.
Speakers	Play sound, music, and videos.
Webcam	Captures photos and videos.
Microphone	Records voice and sound.

Input, Output & Storage (Overview)

Input Devices

Devices used to enter data into the computer.

Examples: Keyboard, Mouse, Scanner, Microphone

Output Devices

Devices used to display or produce information.

Examples: Monitor, Printer, Speakers

Storage Devices

Devices used to store data and files.

Examples: Hard Disk, SSD, Pen Drive, Memory Card

Uses of Computers in Daily Life

Computers are used in many places:

- Schools for learning
 - Homes for education and entertainment
 - Hospitals for patient records
 - Banks for financial transactions
 - Offices for business work
 - Shops for billing
 - Railway stations and airports for ticket booking
 - Libraries for managing books
-

Advantages of Computers

- Fast processing
 - High accuracy
 - Large storage capacity
 - Easy communication
 - Supports online learning
 - Saves time and effort
 - Helps solve problems efficiently
-

Basic Computer Terminology

Term	Meaning
Computer	An electronic device that processes data.
Hardware	Physical parts of a computer.
Software	Programs that tell the computer what to do.
CPU	The brain of the computer.
Monitor	Displays information on the screen.
Keyboard	Used for typing.
Mouse	Used to point, click, and select.
Storage	Saves data and files for future use.

Did You Know?

- The CPU is often called the "**Brain of the Computer.**"
 - Computers can perform millions of calculations in a second.
 - Modern computers are used in almost every profession.
 - Every computer system requires both **hardware and software** to work properly.
-

Points to Remember

- A computer is an electronic device that processes data.
- A computer system consists of hardware, software, users, and data.
- Hardware includes all the physical parts of a computer.
- Internal hardware is inside the computer cabinet, while external hardware is connected from outside.
- Computers are widely used in education, healthcare, banking, business, and communication.
- Proper handling and care help computers work efficiently for a long time.

Chapter 1: Computer Skills

Class 2 – Input, Output & Storage Devices in Daily Life

Learning Objectives

After completing this lesson, students will be able to:

- Understand the meaning of input, output, and storage devices.
 - Identify common input, output, and storage devices.
 - Explain the functions of different computer devices.
 - Classify computer devices into the correct categories.
 - Understand how these devices are used in everyday life.
 - Develop basic computer knowledge through real-life examples.
-

Introduction

A computer works by accepting information, processing it, storing it, and displaying the results. To perform these tasks, a computer uses different types of devices called **Input Devices**, **Output Devices**, and **Storage Devices**.

These devices help us interact with the computer and complete various tasks in schools, homes, offices, hospitals, and banks.

What are Input Devices?

Input devices are hardware devices used to enter data and instructions into a computer.

Examples of Input Devices

- Keyboard
- Mouse
- Scanner
- Microphone
- Webcam
- Joystick
- Touchscreen
- Barcode Scanner

Keyboard

A **keyboard** is the most common input device.

Uses

- Typing letters and numbers
- Writing documents
- Entering commands
- Using shortcut keys

Mouse

A **mouse** is a pointing device used to control the cursor.

Uses

- Clicking
- Double-clicking
- Dragging and dropping
- Selecting files and folders
- Drawing in Paint

Scanner

A **scanner** converts printed documents and photographs into digital form.

Uses

- Scanning documents
- Saving certificates
- Digitizing photographs

Microphone

A **microphone** is used to record sound.

Uses

- Voice recording
 - Online classes
 - Video calls
 - Voice commands
-

Webcam

A **webcam** captures photos and videos.

Uses

- Online meetings
 - Video conferencing
 - Online learning
 - Recording videos
-

What are Output Devices?

Output devices display or produce the results processed by the computer.

Examples of Output Devices

- Monitor
 - Printer
 - Speakers
 - Projector
 - Headphones
-

Monitor

A **monitor** displays text, pictures, videos, and other information.

Uses

- Watching videos
- Reading documents

- Viewing presentations
 - Playing educational games
-

Printer

A **printer** produces a paper copy (hard copy) of digital documents.

Uses

- Printing assignments
 - Printing photographs
 - Printing certificates
 - Printing reports
-

Speakers

Speakers produce sound from the computer.

Uses

- Listening to music
 - Watching videos
 - Online learning
 - Educational audio lessons
-

Projector

A **projector** displays the computer screen on a large surface.

Uses

- Classroom teaching
 - Business presentations
 - Seminars
 - Training sessions
-

What are Storage Devices?

Storage devices are used to save data, files, pictures, videos, and documents for future use.

Examples of Storage Devices

- Hard Disk (HDD)
 - Solid State Drive (SSD)
 - Pen Drive
 - Memory Card
 - CD/DVD
 - External Hard Disk
-

Hard Disk (HDD)

The **Hard Disk** is the main storage device inside a computer.

Uses

- Stores operating system
 - Saves software
 - Stores files and folders
 - Keeps documents permanently
-

Solid State Drive (SSD)

An **SSD** stores data like a hard disk but works much faster.

Uses

- Faster computer performance
 - Faster file access
 - Quick system startup
-

Pen Drive

A **Pen Drive** is a portable storage device.

Uses

- Carrying files
 - Sharing documents
 - Transferring data
 - Backup storage
-

Memory Card

A **Memory Card** stores data in mobile phones, cameras, and tablets.

Uses

- Saving photos
 - Storing videos
 - Keeping music files
 - Expanding storage
-

CD/DVD

A **CD/DVD** is an optical storage device.

Uses

- Storing music
 - Saving videos
 - Installing software
 - Data backup
-

Device Classification

Input Devices Output Devices Storage Devices

Keyboard	Monitor	Hard Disk
Mouse	Printer	SSD
Scanner	Speakers	Pen Drive
Microphone	Projector	Memory Card

Input Devices Output Devices Storage Devices

Webcam

Headphones

CD/DVD

Uses of Computer Devices in Daily Life

Computers and their devices are used in many places:

Schools

- Online learning
- Smart classrooms
- Computer labs
- Project presentations

Homes

- Homework
- Entertainment
- Online shopping
- Video calls

Hospitals

- Patient records
- Medical reports
- Scanning documents

Banks

- Money transactions
- Printing receipts
- Customer records

Offices

- Typing documents
 - Printing reports
 - Online meetings
 - Data storage
-

Importance of Input, Output & Storage Devices

These devices help us to:

- Enter information into the computer.
 - View and hear computer output.
 - Save important files safely.
 - Share information easily.
 - Complete tasks quickly and accurately.
-

Basic Computer Terminology

Term	Meaning
Input Device	A device used to enter data into a computer.
Output Device	A device that displays or produces results.
Storage Device	A device used to save data for future use.
Keyboard	Used for typing.
Mouse	Used to control the pointer.
Monitor	Displays information on the screen.
Printer	Prints documents on paper.
Hard Disk	Stores data permanently.

Did You Know?

- A keyboard has over **100 keys** on most standard keyboards.
 - An SSD is faster than a traditional hard disk.
 - A pen drive is small enough to fit in your pocket but can store many files.
 - Projectors help teachers display lessons to the whole class.
-

Points to Remember

- **Input devices** are used to enter data into a computer.
- **Output devices** display or produce the results.

- **Storage devices** save files and information for future use.
- Every computer uses different hardware devices to perform different tasks.
- Proper care of computer devices increases their lifespan.
- Computers and their devices are used in many areas of daily life.

Chapter 1: Computer Skills

Class 3 – Computer Care, Digital Ethics & Responsible Technology Use

Learning Objectives

After completing this lesson, students will be able to:

- Understand the importance of taking care of computers.
 - Follow basic computer lab rules and safety practices.
 - Learn the meaning of digital ethics and responsible technology use.
 - Protect personal information while using digital devices.
 - Create strong passwords and practice safe internet habits.
 - Develop good digital citizenship and responsible online behavior.
-

Introduction

Computers are valuable electronic devices that help us learn, communicate, create, and solve problems. Just like books and school equipment, computers must be handled carefully. As technology becomes part of our daily lives, it is also important to use computers and the internet safely, responsibly, and respectfully.

Good computer care and responsible digital behavior help protect both the user and the computer.

Computer Care and Maintenance

Computer care means keeping the computer clean, safe, and working properly.

Good Computer Care Practices

- Keep the computer clean and dust-free.
 - Wash and dry your hands before using the computer.
 - Use the keyboard and mouse gently.
 - Keep food and drinks away from the computer.
 - Shut down the computer properly after use.
 - Keep cables neat and avoid pulling them.
 - Report any problems to the teacher or lab assistant.
 - Do not hit or shake computer devices.
-

Safe Handling of Computer Equipment

Always handle computer equipment with care.

Safety Rules

- Carry laptops with both hands.
 - Do not touch wires or plugs unnecessarily.
 - Never disconnect cables without permission.
 - Do not press keys too hard.
 - Keep liquids away from electronic devices.
 - Use clean hands while operating the computer.
 - Sit properly while using the computer.
-

Computer Lab Rules

Follow these rules in the computer lab:

- Enter the lab quietly.
 - Follow the teacher's instructions.
 - Use only your assigned computer.
 - Do not change computer settings.
 - Do not install software without permission.
 - Keep the workspace clean.
 - Save your work regularly.
 - Shut down the computer correctly before leaving.
-

What is Digital Ethics?

Digital ethics means using computers, mobile devices, and the internet in a responsible, honest, and respectful way.

Examples of Good Digital Ethics

- Be polite while communicating online.
 - Respect others' opinions.
 - Do not copy another person's work.
 - Ask permission before using someone else's device.
 - Give credit when using another person's ideas or images.
 - Use technology only for learning and positive purposes.
-

Responsible Use of Technology

Technology should be used wisely and responsibly.

Good Technology Habits

- Use computers for learning and creativity.
 - Balance screen time with outdoor activities.
 - Follow school rules while using technology.
 - Do not misuse school computers.
 - Respect digital devices and school property.
 - Use educational websites whenever possible.
-

Internet Safety Basics

The internet is useful, but we must use it safely.

Internet Safety Rules

- Visit only trusted websites.
- Never share personal information online.
- Do not chat with strangers.
- Avoid clicking unknown links.
- Ask an adult before downloading files.
- Inform your teacher or parents if you see anything inappropriate online.

Creating Strong Passwords

A password protects your digital accounts.

Tips for a Strong Password

- Use at least **8 characters**.
- Include uppercase and lowercase letters.
- Include numbers.
- Include special symbols (if allowed).
- Do not use your name or birthday.
- Never share your password with friends.

Example

Strong Password: Learn@2026

Protecting Personal Information

Personal information should always remain private.

Examples of Personal Information

- Full Name
- Home Address
- Phone Number
- School Name
- Passwords
- Bank Information
- Personal Photographs

Safety Tips

- Share personal information only with trusted adults.
 - Never post private information online.
 - Think carefully before sharing photos or videos.
-

Cyber Safety Rules

Cyber safety means staying safe while using the internet.

Remember to

- Think before you click.
 - Be kind and respectful online.
 - Report cyberbullying immediately.
 - Do not open suspicious emails or messages.
 - Log out after using shared computers.
 - Keep your passwords secure.
-

Respecting Others Online

Treat others online just as you would in real life.

Good Online Behaviour

- Use polite language.
 - Respect different opinions.
 - Do not send hurtful messages.
 - Encourage positive communication.
 - Help others use technology safely.
-

Screen Time Awareness

Too much screen time can affect our health.

Healthy Screen Habits

- Take a short break every 20–30 minutes.
 - Sit at a proper distance from the monitor.
 - Maintain good posture.
 - Spend time reading, exercising, and playing outdoors.
 - Avoid using digital devices before bedtime for long periods.
-

Good Digital Habits

Develop these habits every day:

- Save your work regularly.
 - Keep files organized.
 - Log out after using shared devices.
 - Use technology responsibly.
 - Follow classroom and lab rules.
 - Respect school equipment.
-

What is a Digital Footprint?

A **digital footprint** is the information we leave behind when we use the internet or digital devices.

Examples

- Sending emails
- Posting photos
- Watching videos
- Playing online games
- Visiting websites

Always remember that what we do online can leave a lasting record.

Importance of Computer Care & Digital Ethics

Learning these skills helps students:

- Protect computers from damage.
- Stay safe online.
- Use technology responsibly.
- Respect others while using digital devices.
- Develop good digital citizenship.
- Become responsible digital learners.

Basic Computer Terminology

Term	Meaning
Computer Care	Keeping a computer clean and safe.
Digital Ethics	Responsible and respectful use of technology.
Internet Safety	Safe use of the internet.
Password	A secret code used to protect accounts.
Personal Information	Private details about a person.
Cyber Safety	Staying safe while using digital technology.
Digital Citizenship	Responsible behavior while using technology.
Digital Footprint	The record of activities done online.

Did You Know?

- Good digital citizens use technology safely and respectfully.
 - Strong passwords help protect personal information.
 - A clean computer works better and lasts longer.
 - Everything shared online can become part of your digital footprint.
-

Points to Remember

- Handle computers carefully.
- Follow all computer lab rules.
- Be polite and respectful online.
- Keep your personal information private.
- Create strong passwords and never share them.
- Use the internet safely and responsibly.
- Balance screen time with healthy daily activities.
- Think before you click or share anything online.

Chapter 1: Computer Skills

Class 4 – Revision, Discussion & Assessment

Learning Objectives

After completing this lesson, students will be able to:

- Revise all the concepts learned in Chapter 1.
 - Identify and explain computer hardware devices.
 - Classify input, output, and storage devices correctly.
 - Demonstrate knowledge of computer care and digital ethics.
 - Apply responsible technology use in daily life.
 - Complete chapter assessment activities confidently.
-

Introduction

This lesson is a **revision and assessment session** designed to reinforce all the concepts learned in **Chapter 1: Computer Skills**. Students will review computer fundamentals, hardware devices, digital ethics, and computer care through interactive discussions, revision exercises, and assessments.

Revision of Computer Fundamentals

Students revise:

- What is a Computer?
- Computer System
- Hardware
- Software
- Functions of a Computer
- Uses of Computers in Daily Life

Examples

- Computer
- Hardware
- Software
- CPU
- Monitor
- Keyboard

Revision of Hardware Devices

Students revise the main hardware devices.

Input Devices

- Keyboard
- Mouse
- Scanner
- Microphone
- Webcam

Output Devices

- Monitor
- Printer
- Speakers
- Projector

Storage Devices

- Hard Disk
- SSD
- Pen Drive
- Memory Card
- CD/DVD

Revision of Device Functions

Students review the function of each device.

Device	Function
Keyboard	Types text and numbers
Mouse	Controls the pointer
Monitor	Displays information
Printer	Prints documents
Scanner	Scans documents
Speakers	Produce sound

Device	Function
Pen Drive	Stores files
Hard Disk	Stores data permanently

Revision of Computer Care

Students revise:

- Keeping computers clean
 - Proper handling of devices
 - Computer lab rules
 - Safe shutdown procedures
 - Correct sitting posture
 - Keeping food and drinks away from computers
-

Revision of Digital Ethics

Students revise:

- Responsible use of technology
 - Respecting others online
 - Protecting personal information
 - Strong passwords
 - Safe internet browsing
 - Good digital citizenship
-

Discussion Session

The teacher conducts a classroom discussion using questions such as:

- Why are computers important?
- Which hardware device do you use the most?
- How can we protect a computer?
- Why should we use strong passwords?
- How should we behave online?
- Why is digital citizenship important?
- Where do we use computers in daily life?

Students are encouraged to share their ideas and experiences.

Question & Answer Session

Sample Questions

1. What is hardware?
 2. Name four input devices.
 3. Name three output devices.
 4. What is a storage device?
 5. Why should we take care of computers?
 6. What is digital ethics?
 7. What is cyber safety?
 8. What is a digital footprint?
 9. What is responsible technology use?
 10. Give two examples of good digital habits.
-

Chapter Assessment

Students complete a short assessment.

Section A – Multiple Choice Questions

1. Which is an input device?
 - a) Monitor
 - b) Keyboard
 - c) Printer
 - d) Speaker
2. Which device stores data?
 - a) Mouse
 - b) Pen Drive
 - c) Monitor

d) Scanner

3. Which device displays information?

a) Printer

b) Monitor

c) Keyboard

d) Microphone

Section B – Fill in the Blanks

1. A _____ is used for typing.
 2. A _____ stores files permanently.
 3. Never share your _____ with others.
 4. The _____ displays information on the screen.
 5. We should use technology _____.
-

Section C – True or False

1. A printer is an input device. _____
 2. Computers should be handled carefully. _____
 3. A mouse is used to move the pointer. _____
 4. Sharing passwords with friends is safe. _____
 5. Digital citizenship means using technology responsibly. _____
-

Oral Assessment

Students answer simple oral questions individually.

Sample Oral Questions

- What is a computer?
- What is hardware?
- Name one input device.
- Name one output device.
- Why is computer care important?

- What is digital ethics?
 - What is a strong password?
 - How can you stay safe online?
-

Importance of Revision

Revision helps students:

- Remember important concepts.
 - Improve understanding.
 - Build confidence.
 - Prepare for assessments.
 - Strengthen practical knowledge.
-

Did You Know?

- Regular revision improves memory and learning.
 - Responsible technology users help create a safe digital environment.
 - Taking care of computers increases their lifespan.
 - Good digital habits protect both users and devices.
-

New Words

Word	Meaning
Revision	Reviewing previously learned topics.
Assessment	A test to measure learning and understanding.
Hardware	Physical parts of a computer.
Digital Ethics	Responsible use of technology.
Digital Citizenship	Safe and respectful use of digital technology.
Password	A secret code used to protect accounts.
Cyber Safety	Safe use of computers and the internet.
Technology	Tools and devices that help people perform tasks.

Points to Remember

- Revise all chapter topics regularly.
- Identify hardware devices and explain their functions.
- Classify input, output, and storage devices correctly.
- Follow computer lab rules at all times.
- Use strong passwords and protect personal information.
- Practice good digital citizenship and responsible technology use.
- Handle computers carefully and respectfully.